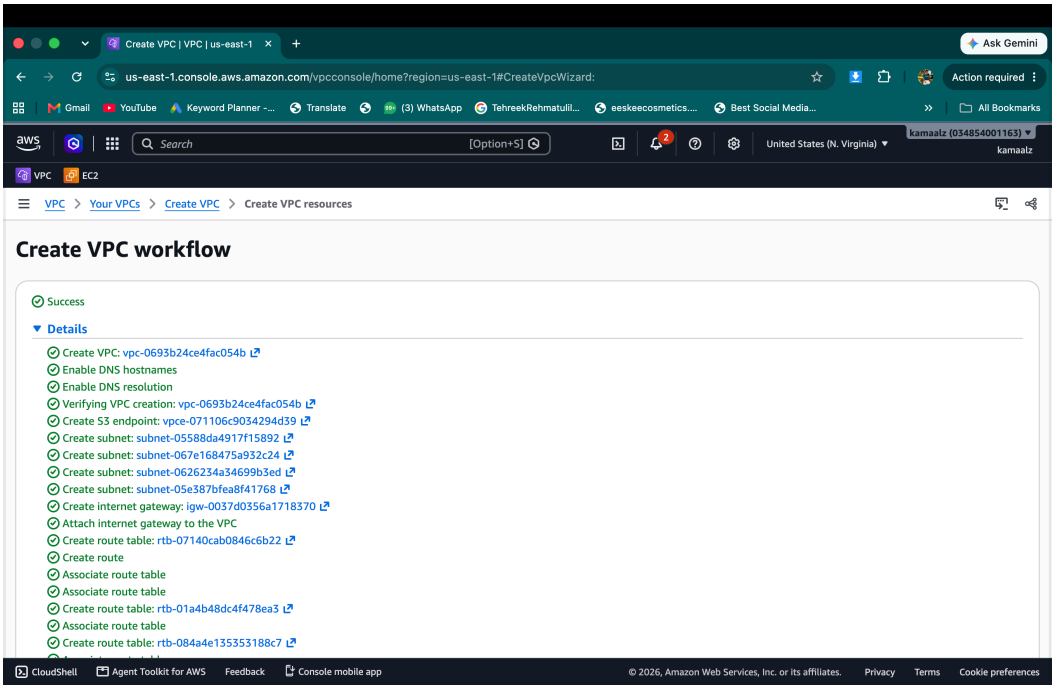
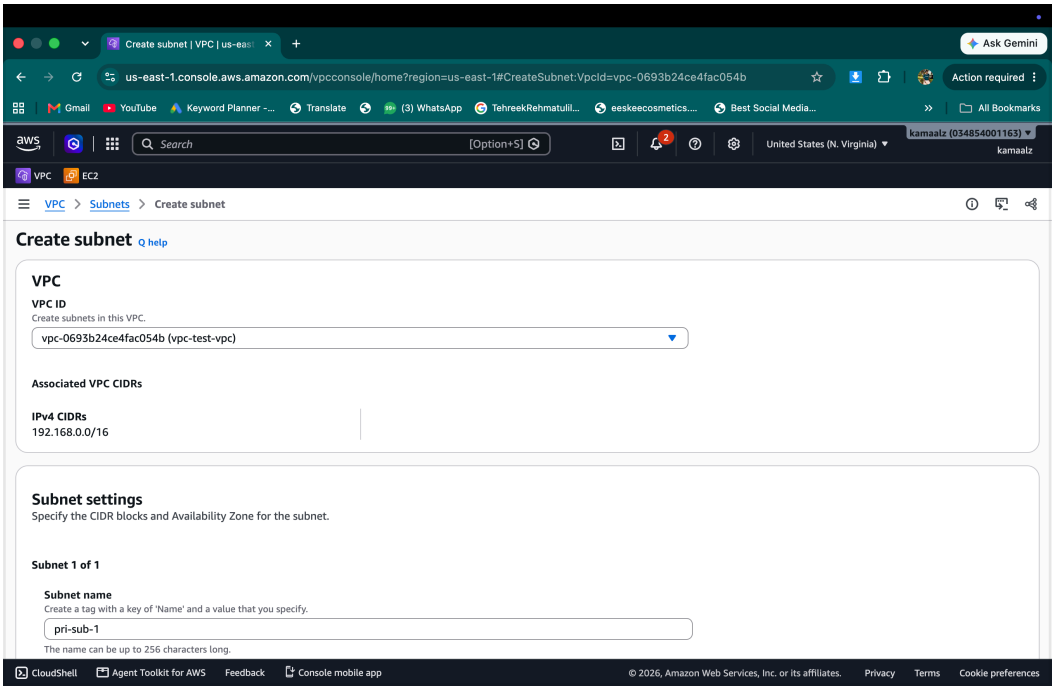
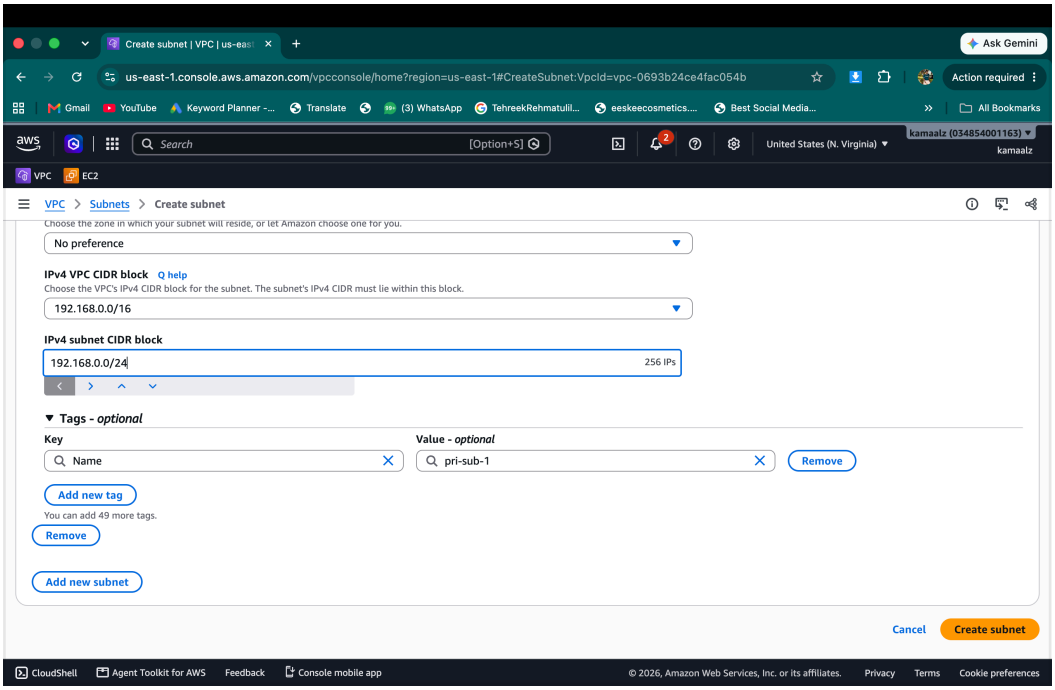
1.Create VPC with 2 private and 2 public subnets.

- Steps to execute:
- • Open the AWS Management Console and navigate to the VPC Dashboard.
- • Select Your VPCs and click Create VPC.
- • Choose VPC only and provide a suitable VPC name.
- • Configure an IPv4 CIDR block, for example 10.0.0.0/16.
- • Navigate to Subnets and create four subnets within the newly created VPC.
- • Create two public subnets in different Availability Zones.
- • Create two private subnets in different Availability Zones.
- • Assign separate CIDR blocks to each subnet without overlapping.
- • Verify that all four subnets are available and associated with the correct VPC.



Create private subnets


Create subnet | VPC | us-east
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateSubnet:VpcId=vpc-0693b24ce4fac054b
Action required
Gmail
YouTube
Keyword Planner -...
Translate
(3) WhatsApp
TehreekRehmatulil...
eeskeecosmetics....
Best Social Media...
All Bookmarks
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
VPC
EC2
VPC > Subnets > Create subnet
Create subnet
Q help
VPC
VPC ID
Create subnets in this VPC.
vpc-0693b24ce4fac054b (vpc-test-vpc)
Associated VPC CIDRs
IPv4 CIDRs
192.168.0.0/16
Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
pri-sub-1
The name can be up to 256 characters long.
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Create subnet | VPC | us-east
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateSubnet:VpcId=vpc-0693b24ce4fac054b
Action required
Gmail
YouTube
Keyword Planner -...
Translate
(3) WhatsApp
TehreekRehmatulil...
eeskeecosmetics....
Best Social Media...
All Bookmarks
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
VPC
EC2
VPC > Subnets > Create subnet
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference
IPv4 VPC CIDR block
Q help
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/16
IPv4 subnet CIDR block
192.168.0.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
pri-sub-1
Remove
Add new tag
You can add 49 more tags.
Remove
Add new subnet
Cancel
Create subnet
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
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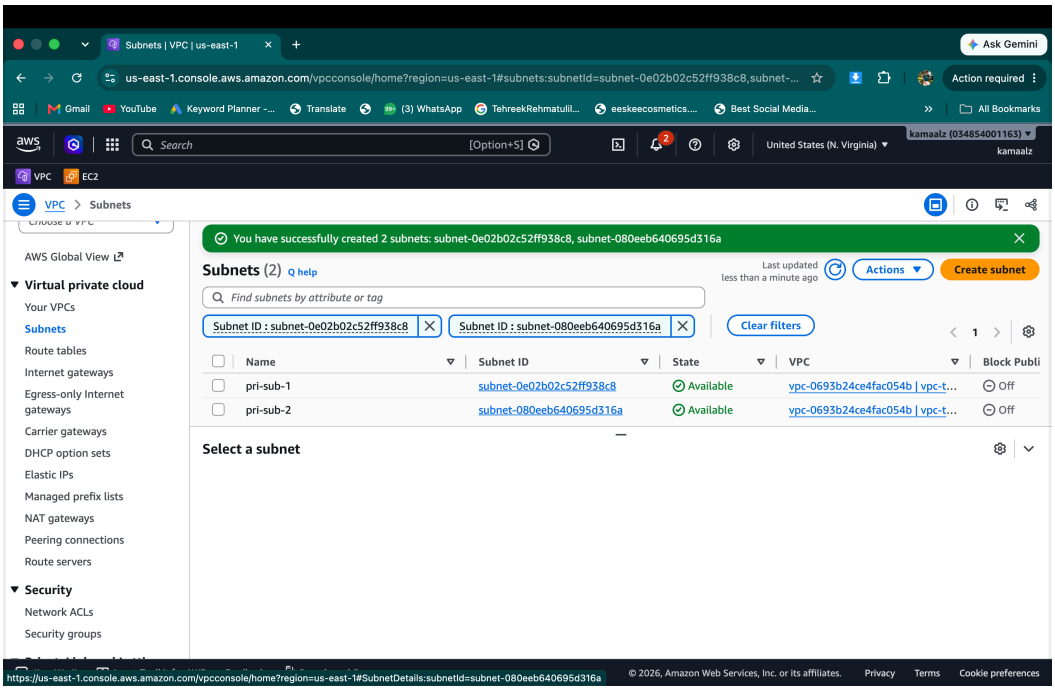
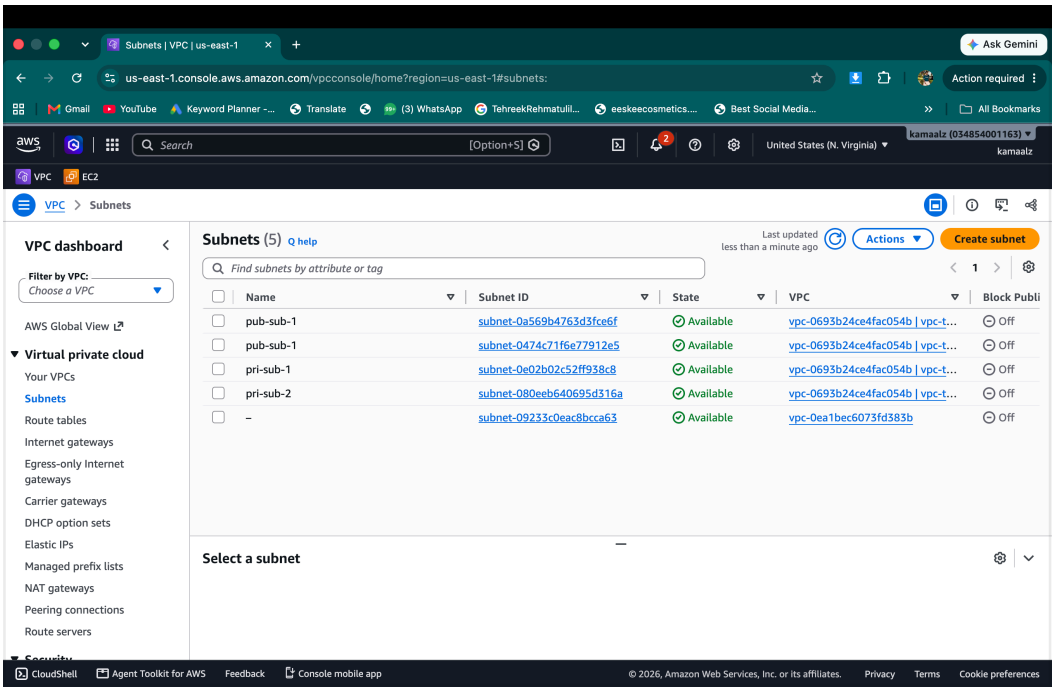


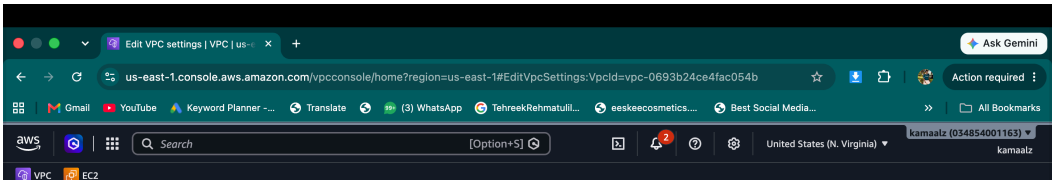
Created 2 private subnets and 2 public subnets



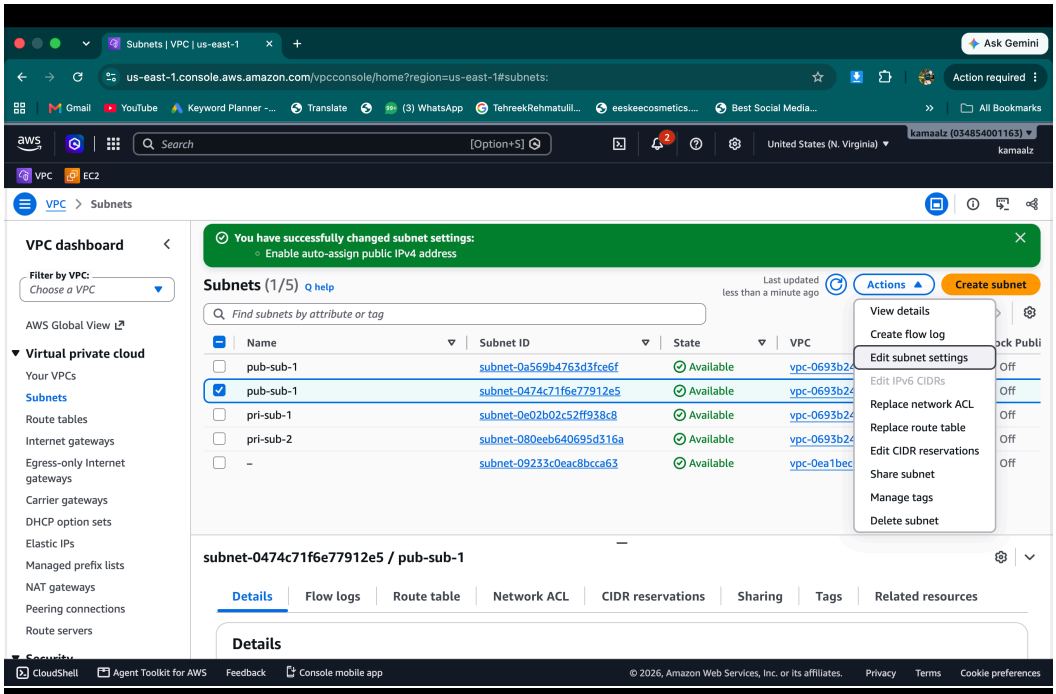
3.Enable DNS Hostname in VPC.

- Steps to execute:
- • Navigate to VPC Dashboard and select Your VPCs.
- • Select the VPC created in Task 1.
- • Click the Actions dropdown.
- • Select Edit VPC settings.
- • Enable the DNS resolution option.
- • Enable the DNS hostnames option.
- • Save the changes.

• Verify that both DNS settings are enabled in the VPC details.



1.


Subnets | VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:
Action required
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
VPC
EC2
VPC > Subnets
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
You have successfully changed subnet settings:
Enable auto-assign public IPv4 address
Subnets (1/5) Q help
Last updated less than a minute ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
pub-sub-1
subnet-0a569b4763d3fce6f
Available
pub-sub-1
subnet-0474c71f6e77912e5
Available
pri-sub-1
subnet-0e02b02c52ff938c8
Available
pri-sub-2
subnet-080eeb640695d316a
Available
–
subnet-09233c0eac8bcca63
Available
View details
Create flow log
Edit subnet settings
Edit IPv6 CIDRs
Replace network ACL
Replace route table
Edit CIDR reservations
Share subnet
Manage tags
Delete subnet
subnet-0474c71f6e77912e5 / pub-sub-1
Details
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Details
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
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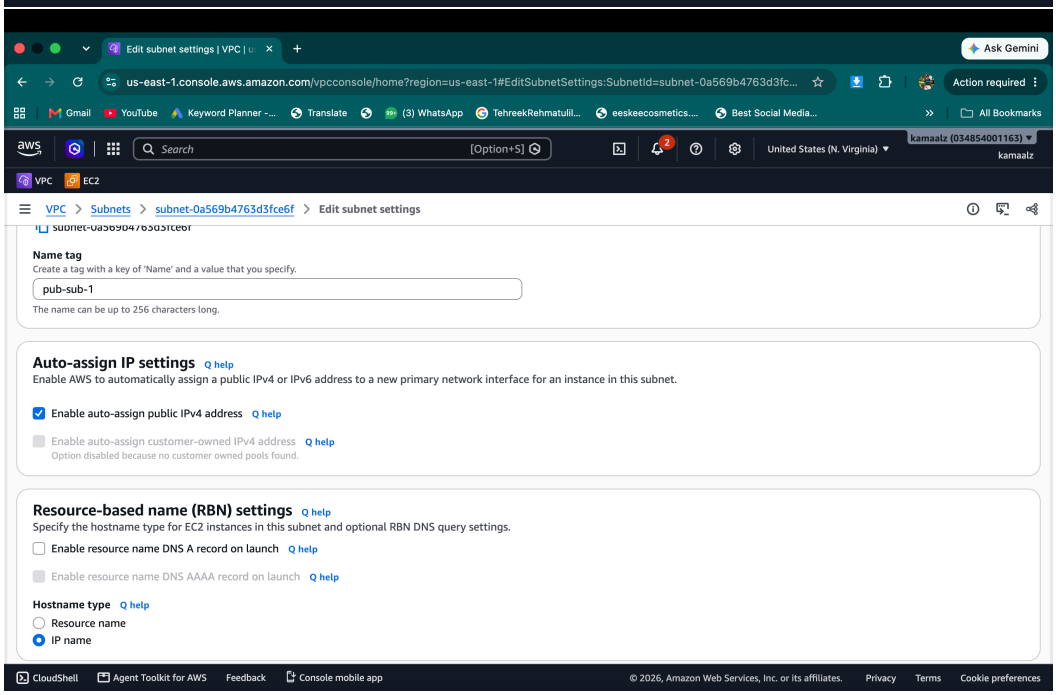
2.


Edit subnet settings | VPC | u
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditSubnetSettings:SubnetId=subnet-0a569b4763d3fc...
Action required
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
VPC
EC2
VPC > Subnets > subnet-0a569b4763d3fce6f > Edit subnet settings
Name tag
Create a tag with a key of 'Name' and a value that you specify.
pub-sub-1
The name can be up to 256 characters long.
Auto-assign IP settings Q help
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.
Enable auto-assign public IPv4 address Q help
Enable auto-assign customer-owned IPv4 address Q help
Option disabled because no customer owned pools found.
Resource-based name (RBN) settings Q help
Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.
Enable resource name DNS A record on launch Q help
Enable resource name DNS AAAA record on launch Q help
Hostname type Q help
Resource name
IP name
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
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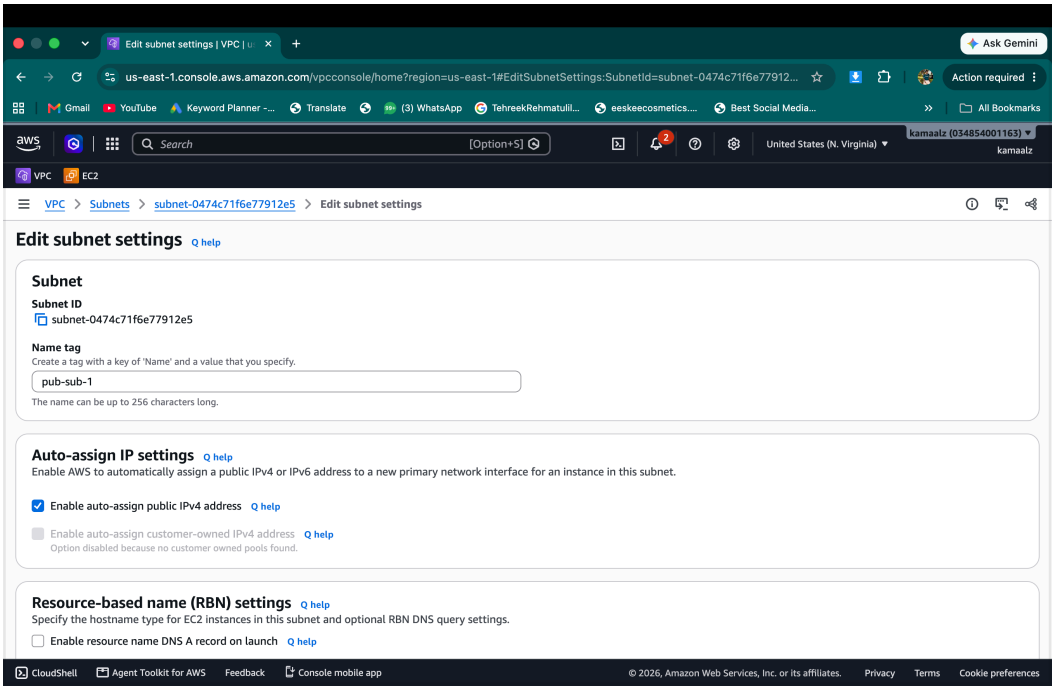
Cookie preferences

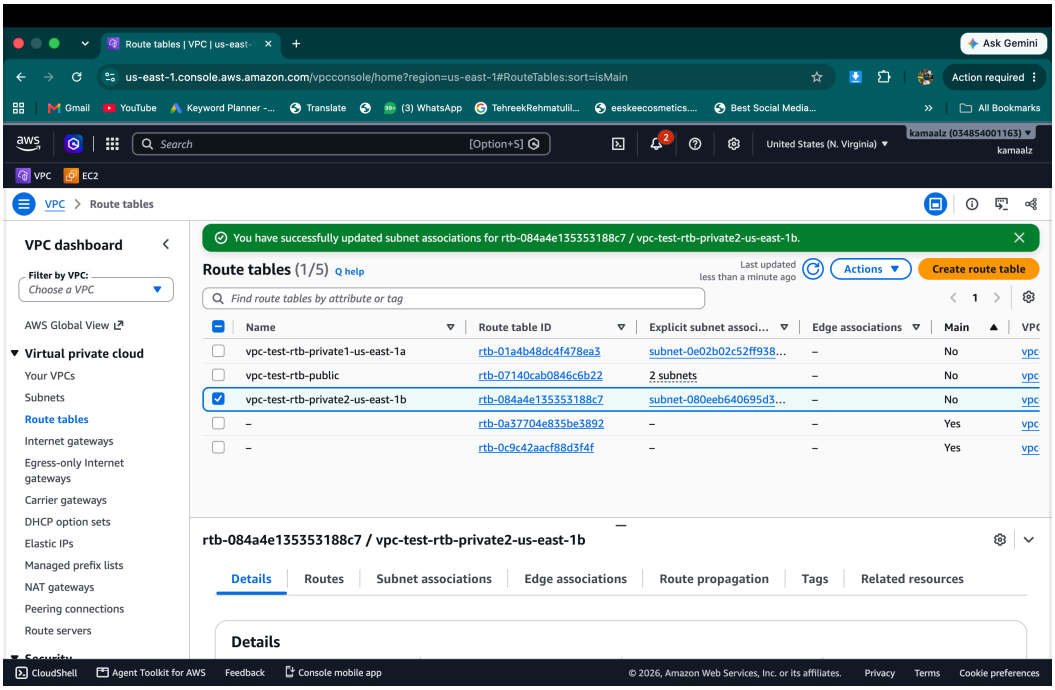


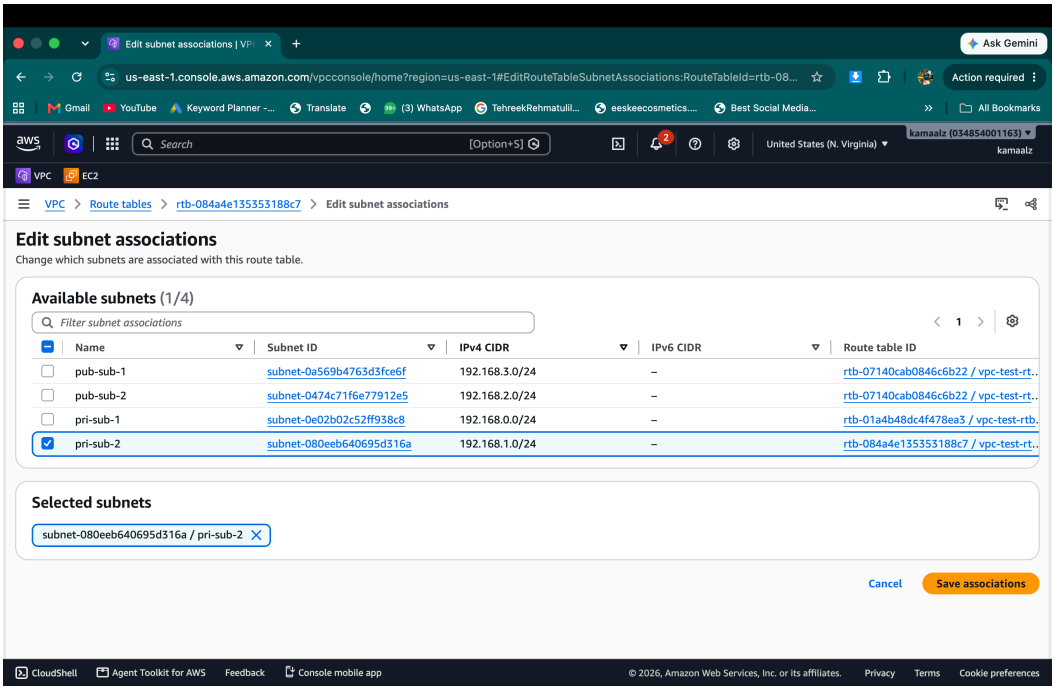
4.Add 2 private subnets in private route table.

- Steps to execute:
- • Navigate to VPC → Route Tables.
- • Click Create route table.
- • Provide a name such as Private-Route-Table.
- • Select the VPC created in Task 1.
- • Create the route table.
- • Select the newly created private route table.
- • Open the Subnet associations tab.
- • Click Edit subnet associations.
- • Select both private subnets.
- • Save the associations.

1. • Verify that both private subnets are associated with the private route table.

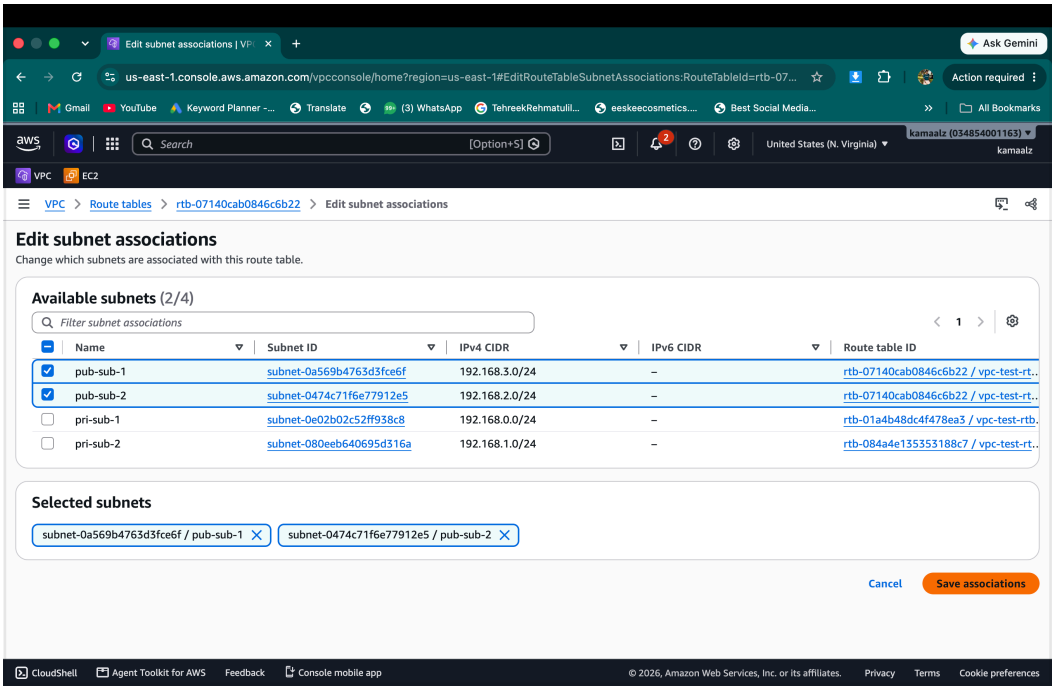
2.





3. Add 2 public subnets in public route table.
   - Steps to execute:
   - • Navigate to VPC → Route Tables.
   - • Click Create route table.
   - • Provide a name such as Public-Route-Table.
   - • Select the VPC created earlier.
   - • Create the route table.
   - • Select the public route table.
   - • Open the Subnet associations tab.
   - • Click Edit subnet associations.
   - • Select both public subnets.
   - • Save the subnet associations.

• Verify that both public subnets are associated with the public route table.

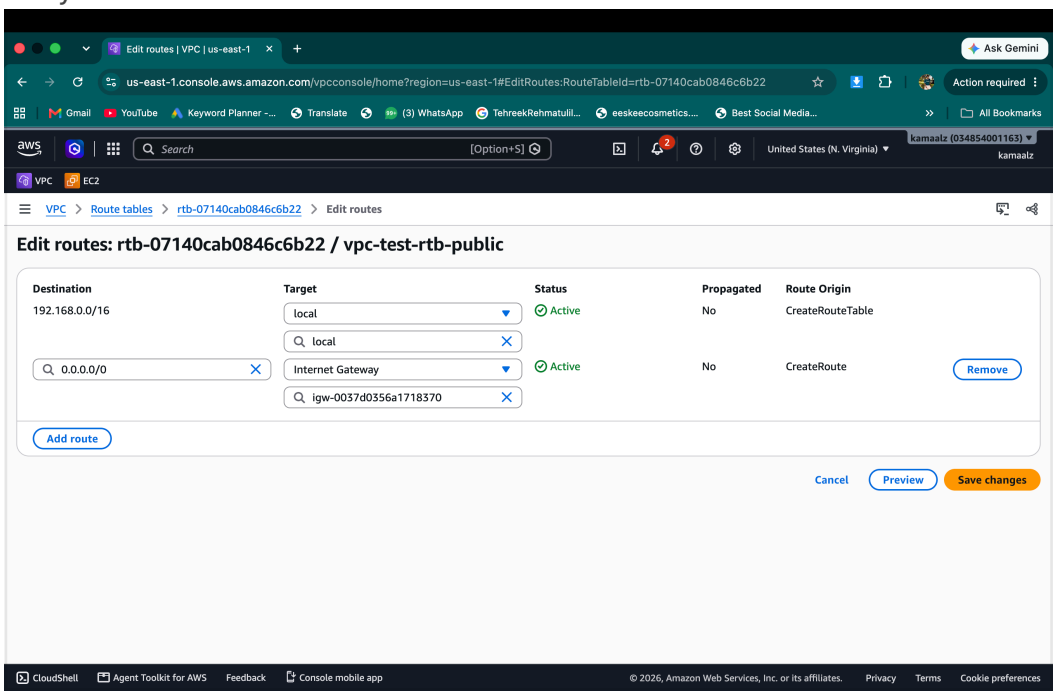


1.

6.Public route table will have the routes to

- internet and local.
- Steps to execute:
- • Navigate to VPC → Internet Gateways.
- • Click Create internet gateway and provide a suitable name.
- • Select the newly created Internet Gateway.
- • Click Actions → Attach to VPC.
- • Select the VPC created in Task 1 and attach the gateway.
- • Navigate to Route Tables and select the public route table.
- • Open the Routes tab and click Edit routes.
- • Add a new route with destination 0.0.0.0/0.
- • Select Internet Gateway as the target and choose the attached Internet Gateway.
- • Save the changes.

• Verify that the route table contains both the local VPC route and the Internet Gateway route.

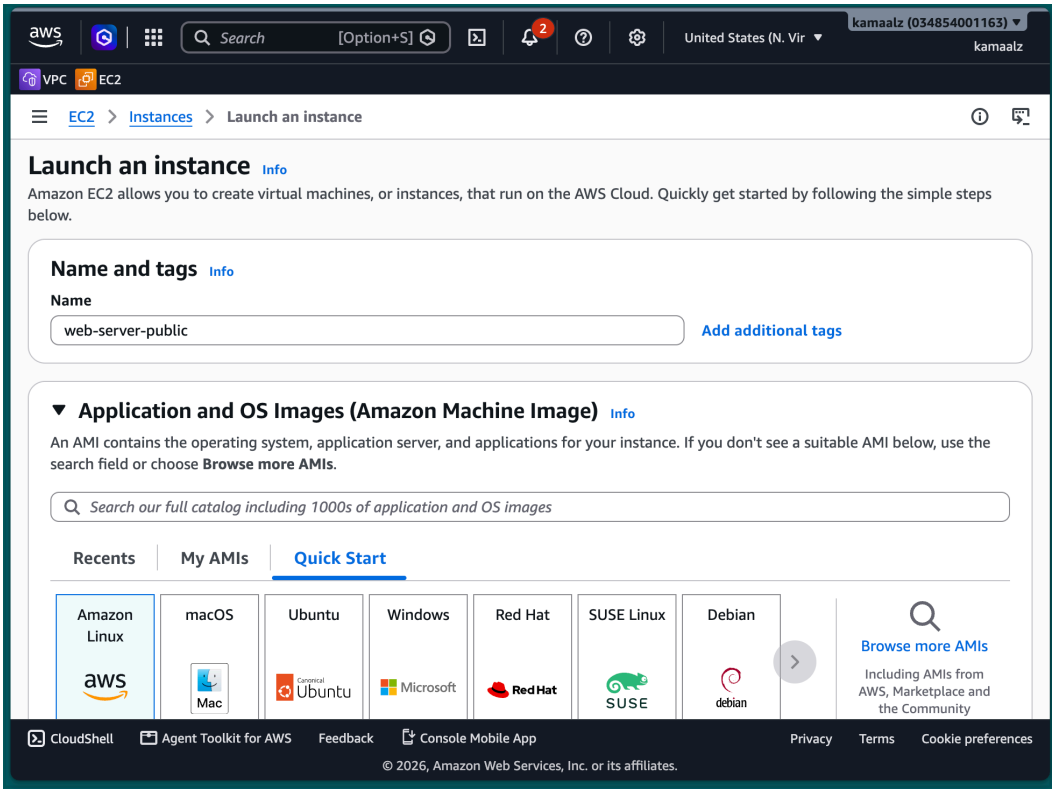
1. 

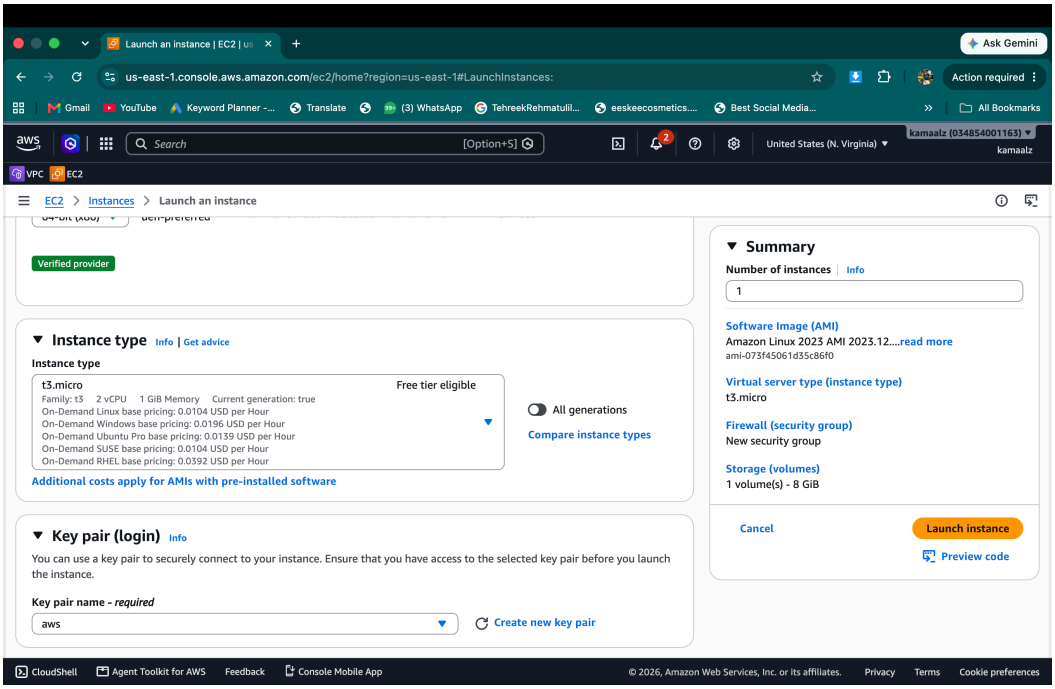
7.Create EC2 in public subnet with t2.micro

- and install PHP.
- Steps to execute:
- • Navigate to EC2 → Instances and click Launch instances.
- • Provide an instance name such as Public-PHP-Server.
- • Select Amazon Linux 2023 as the AMI.
- • Choose the t2.micro instance type.
- • Select an existing key pair or create a new key pair.
- • Under Network settings, select the VPC created in Task 1.
- • Select one of the public subnets.
- • Enable Auto-assign public IP.
- • Configure the security group to allow SSH (port 22) from your IP and HTTP (port 80) if

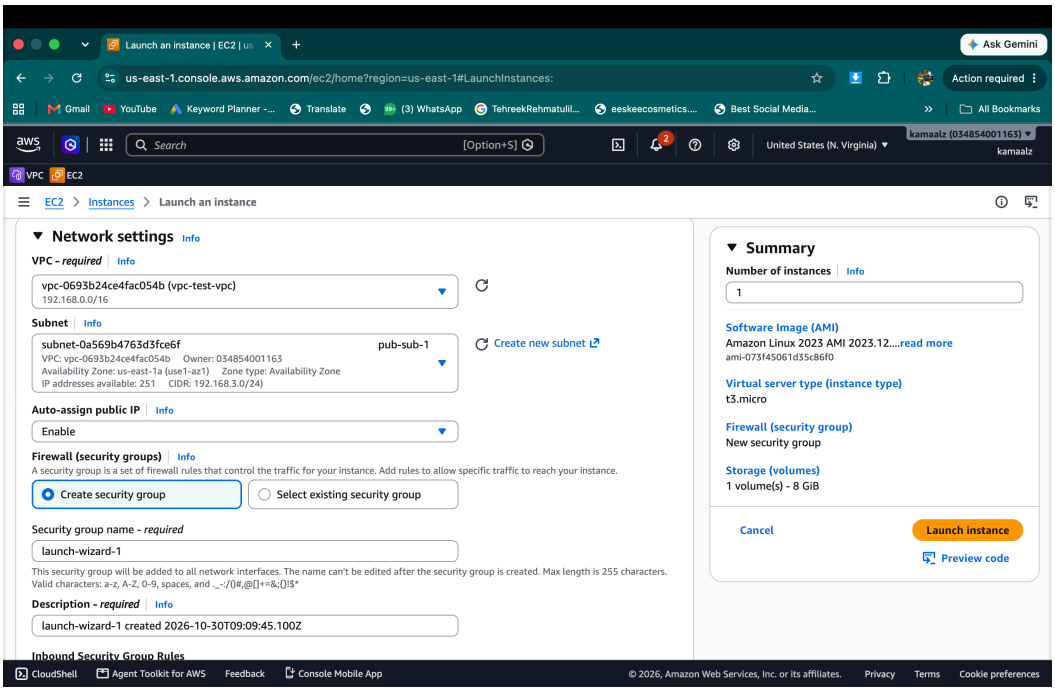
- hosting a PHP webpage.
- • Launch the instance and wait until it reaches the Running state.
- • Connect to the instance using EC2 Instance Connect or SSH.
- • Update the system packages and install Apache and PHP.
- • Start and enable the Apache service.
- • Create a sample PHP webpage in the Apache document root.

• Test the PHP webpage using the public IPv4 address of the EC2 instance

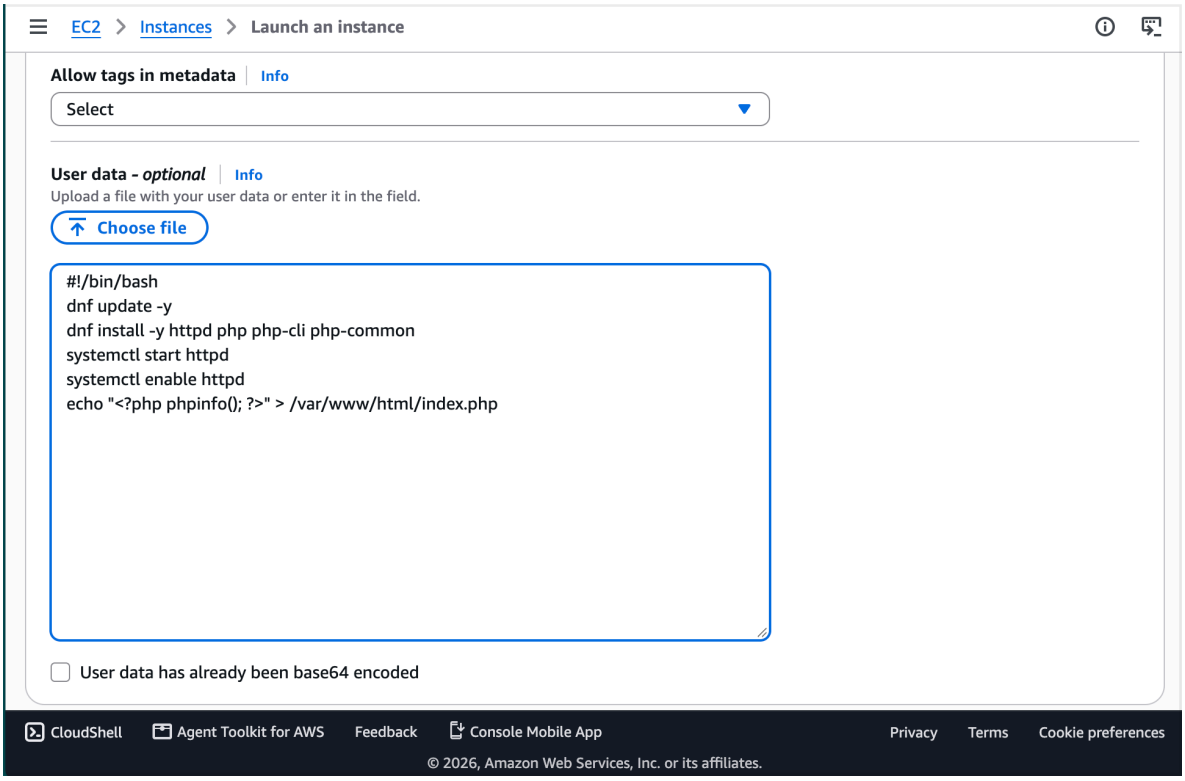
1. 



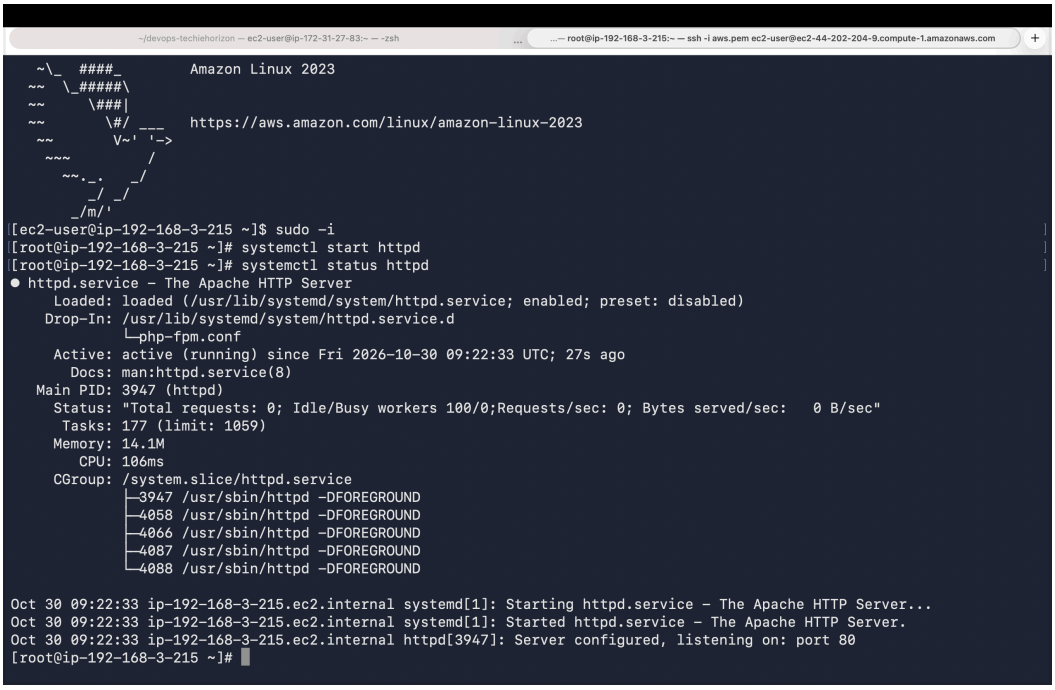
step3:select vpc we created and also subnet



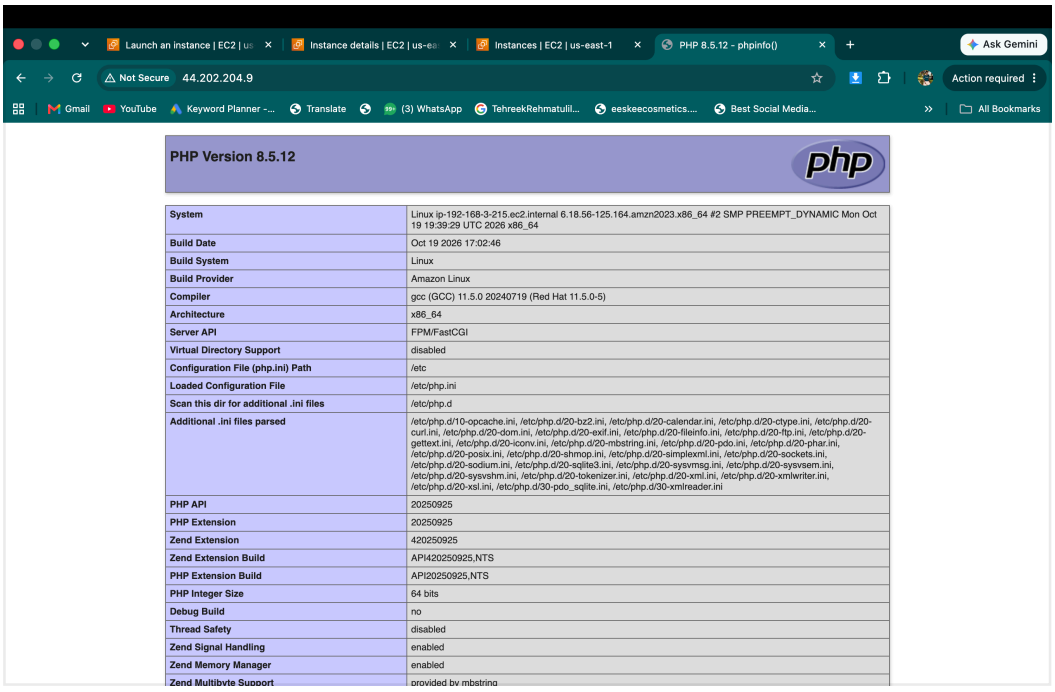
step4:Add this script in advance section
#!/bin/bash
dnf update -y
dnf install -y httpd php php-cli php-common
systemctl start httpd
systemctl enable httpd
echo "<?php phpinfo(); ?>" > /var/www/html/index.php



step 5: connect it local server and check it and by using this command systemctl start httpd then systemctl status httpd



step6:check it on web server using public ip

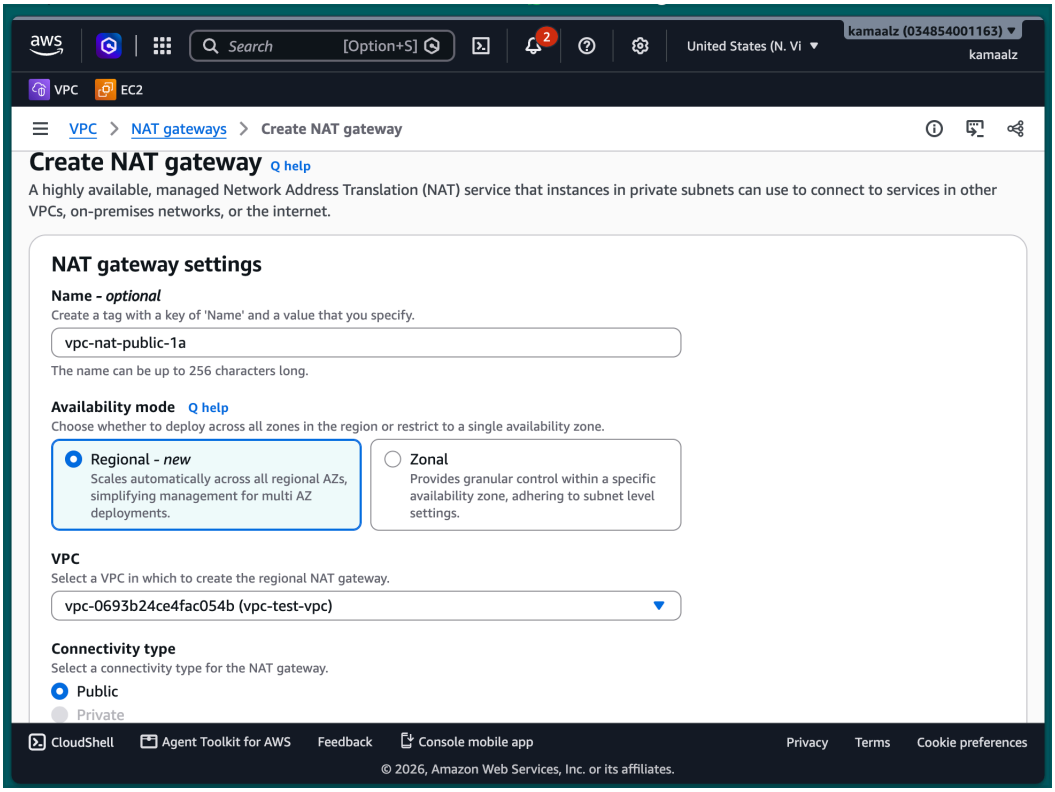


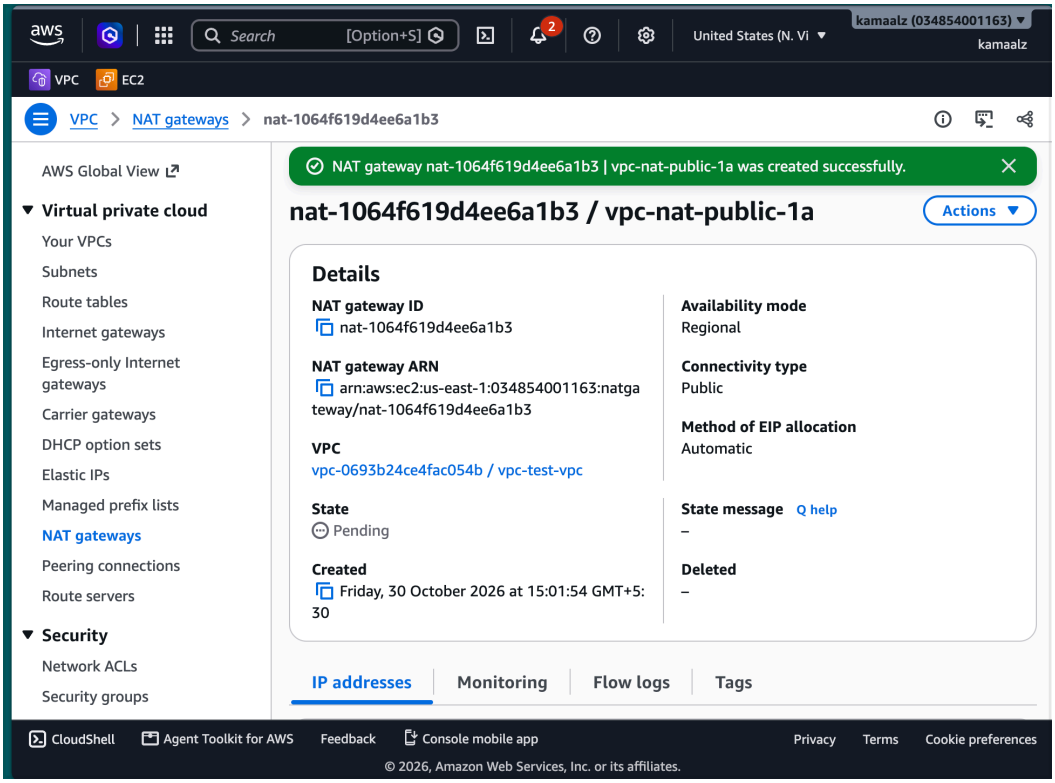
8.Configure NAT gateway in public subnet

- and connect to private instance.
- Steps to execute:
- • Navigate to VPC → NAT Gateways.
- • Click Create NAT Gateway.
- • Provide a suitable name for the NAT Gateway.
- • Select the VPC created in Task 1.
- • Select one of the public subnets.
- • Allocate a new Elastic IP address and associate it with the NAT

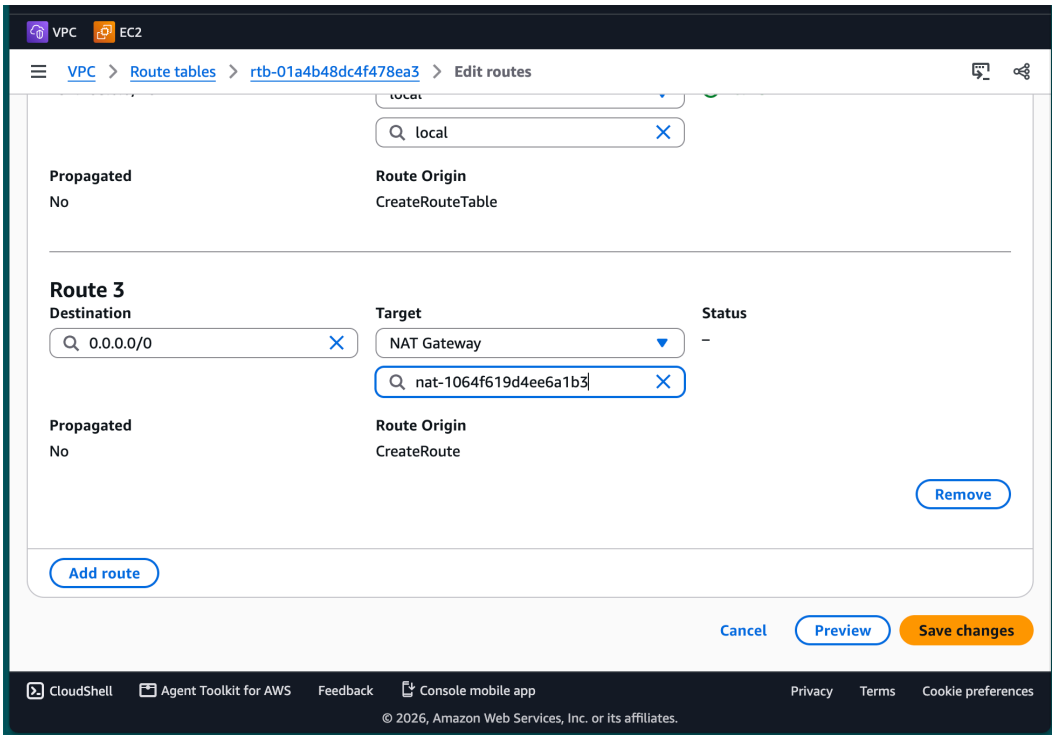
Gateway.

- • Create the NAT Gateway and wait until its state changes to Available.
- • Navigate to Route Tables and select the private route table.
- • Open the Routes tab and click Edit routes.
- • Add a route with destination 0.0.0.0/0.
- • Select the newly created NAT Gateway as the target.
- • Save the route configuration.
- • Launch a private EC2 instance using one of the private subnets.
- • Disable public IP assignment for the private instance.
- • Configure the security group to allow SSH from the public EC2 instance's private IP or security
- group, as appropriate.
- • Connect to the private instance through the public EC2 instance using its private IP.
- • Verify outbound internet connectivity from the private instance through the NAT Gateway.

1. 



step2: next we got to vpc-routetable and add route and connect nat gateway to private subnet

1. 

step3:create a private ec2 and connect private subnet and **Auto-assign public IP to be** disable

Network settings Info

VPC - *required* Info

vpc-0693b24ce4fac054b (vpc-test-vpc)
192.168.0.0/16

Subnet Info

subnet-0e02b02c52ff938c8 pri-sub-1
VPC: vpc-0693b24ce4fac054b Owner: 034854001163
Availability Zone: us-east-1a (use1-az1) Zone type: Availability Zone
IP addresses available: 250 CIDR: 192.168.0.0/24)

Create new subnet

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

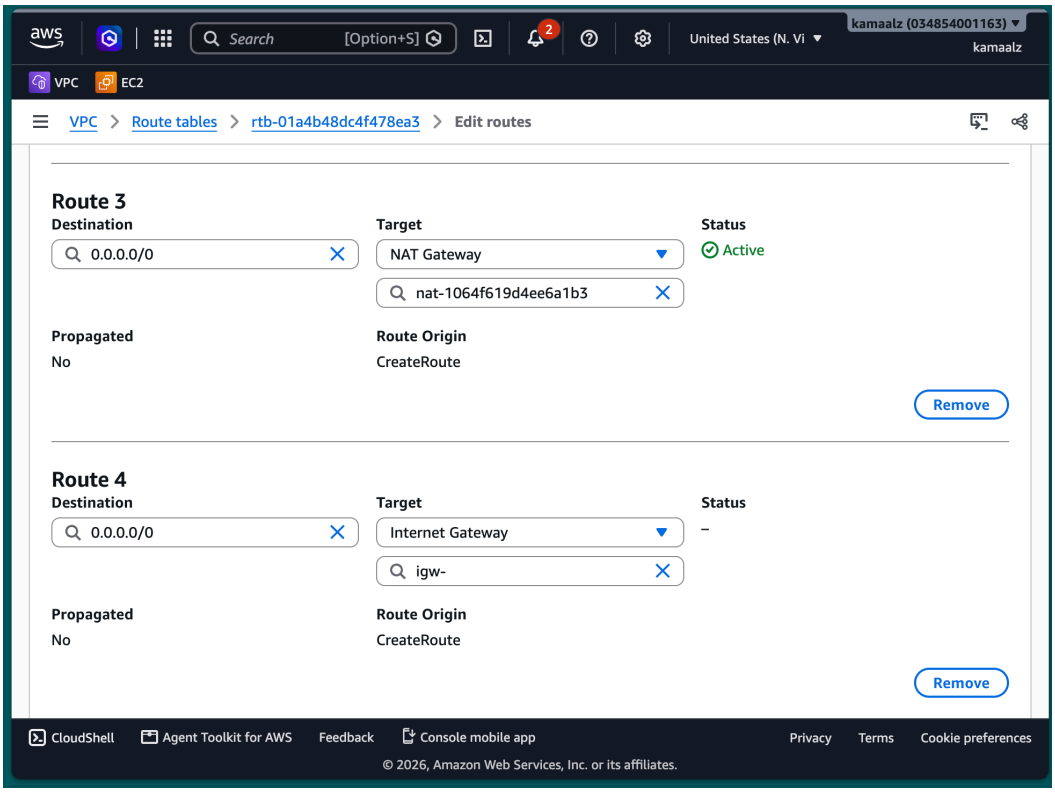
Security group name - *required*

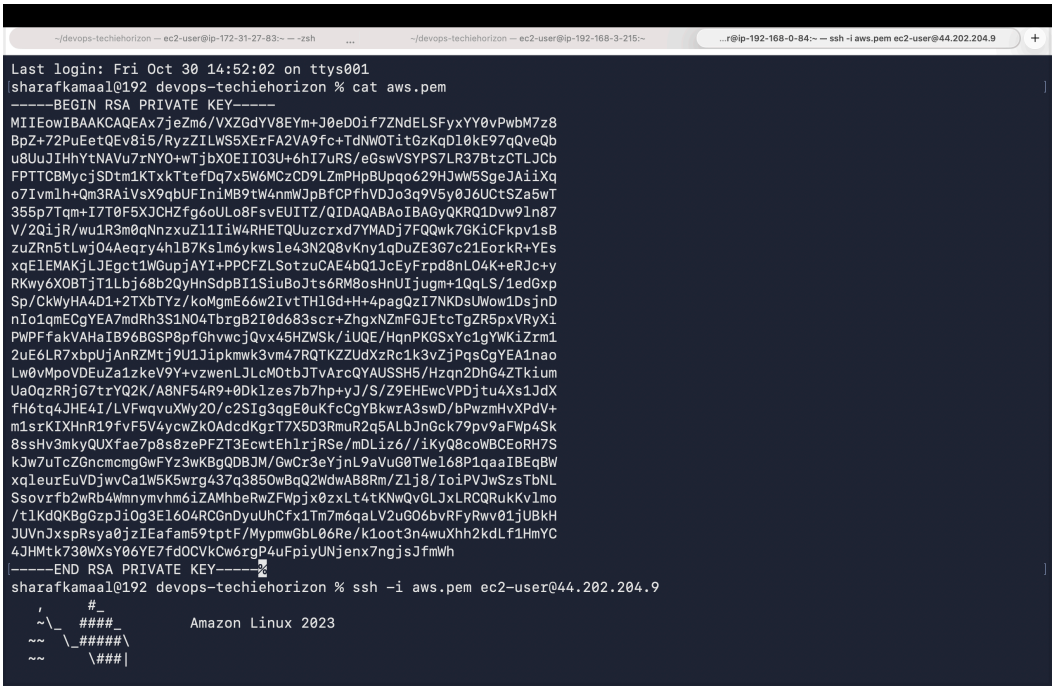
launch-wizard-2

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* Info

launch-wizard-2 created 2026-10-30T09:44:58.664Z

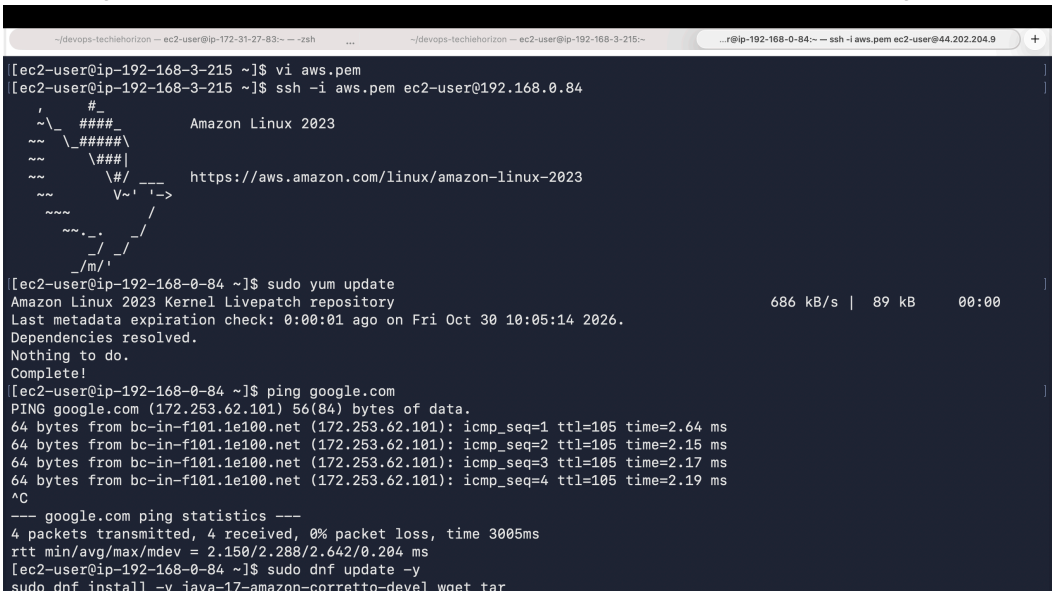





```
sharafkamaal@192 devops-techiehorizon % ssh -i aws.pem ec2-user@44.202.204.9
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 30 09:58:41 2026 from 103.203.172.230
[ec2-user@ip-192-168-3-215 ~]$ ls
aws.pem
[ec2-user@ip-192-168-3-215 ~]$ vi aws.pem
[ec2-user@ip-192-168-3-215 ~]$ ssh -i "aws.pem" ec2-user@192.168.0.84
Load key "aws.pem": invalid format
ec2-user@192.168.0.84: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-192-168-3-215 ~]$ ssh -i "aws.pem" ec2-user@192.168.0.84
Load key "aws.pem": invalid format
ec2-user@192.168.0.84: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-192-168-3-215 ~]$ ssh -i aws.pem ec2-user@192.168.0.84
Load key "aws.pem": invalid format
ec2-user@192.168.0.84: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-192-168-3-215 ~]$ ls
aws.pem
[ec2-user@ip-192-168-3-215 ~]$ vi aws.pem
[ec2-user@ip-192-168-3-215 ~]$ ssh -i aws.pem ec2-user@192.168.0.84
   ,     #_
```

```
[ec2-user@ip-192-168-3-215 ~]$ vi aws.pem
[ec2-user@ip-192-168-3-215 ~]$ ssh -i aws.pem ec2-user@192.168.0.84
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-192-168-0-84 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository                                    686 kB/s |  89 kB     00:00
Last metadata expiration check: 0:00:01 ago on Fri Oct 30 10:05:14 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-192-168-0-84 ~]$ ping google.com
PING google.com (172.253.62.101) 56(84) bytes of data.
64 bytes from bc-in-f101.1e100.net (172.253.62.101): icmp_seq=1 ttl=105 time=2.64 ms
64 bytes from bc-in-f101.1e100.net (172.253.62.101): icmp_seq=2 ttl=105 time=2.15 ms
64 bytes from bc-in-f101.1e100.net (172.253.62.101): icmp_seq=3 ttl=105 time=2.17 ms
64 bytes from bc-in-f101.1e100.net (172.253.62.101): icmp_seq=4 ttl=105 time=2.19 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 2.150/2.288/2.642/0.204 ms
[ec2-user@ip-192-168-0-84 ~]$
```

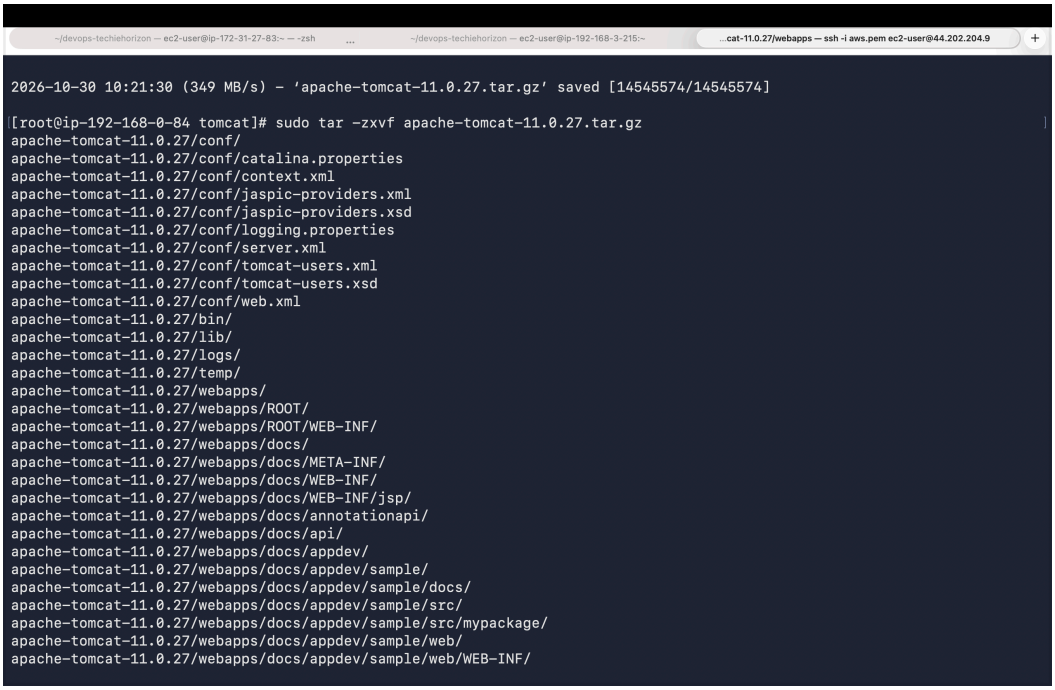
9. Install Apache Tomcat in private EC2 and
   - deploy a sample app.
   - Steps to execute:
   - • Connect to the public EC2 instance.
   - • Use the private IP address of the private EC2 instance to connect to it.
   - • Update the packages on the private EC2 instance.
   - • Install the required Java version and verify the installation.
   - • Download the Apache Tomcat installation package.
   - • Extract the Tomcat archive and configure the required permissions.
   - • Start the Tomcat server and verify that the service is running.
   - • Create a sample HTML application containing a title and application details.
   - • Place the sample application inside the Tomcat webapps directory.
   - • Configure the required security group rules to allow communication from the public EC2
   - instance to the private EC2 instance on port 8080.
   - • From the public EC2 instance, test the private Tomcat application using its private IP address
   - and port 8080.
   1. • Verify that the sample application responds successfully



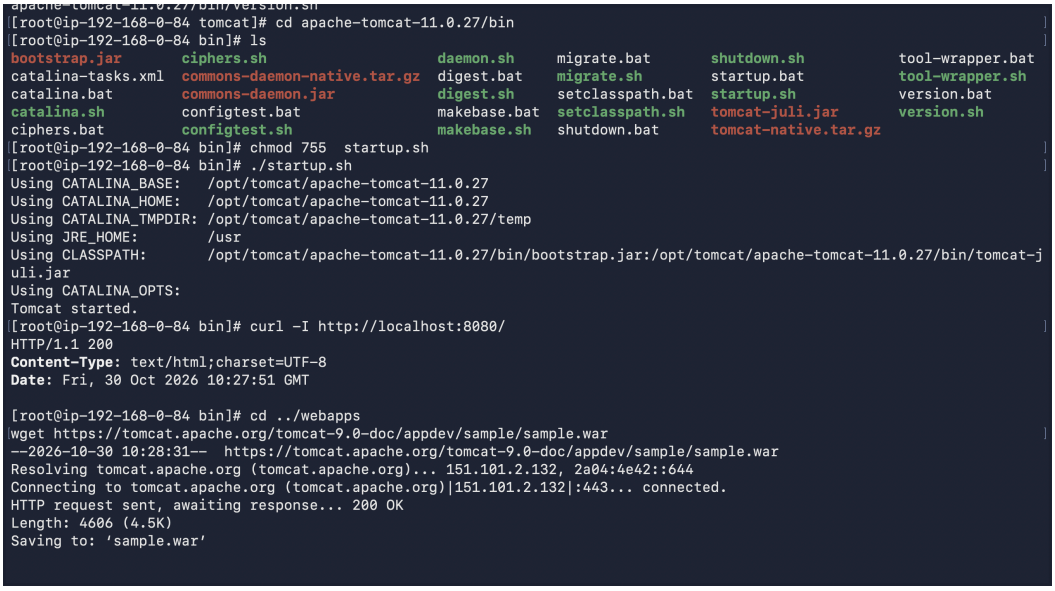
```
[ec2-user@ip-192-168-0-84 ~]$ sudo dnf update -y
sudo dnf install -y java-17-amazon-corretto-devel wget tar
java -version
Last metadata expiration check: 0:07:34 ago on Fri Oct 30 10:05:14 2026.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:07:36 ago on Fri Oct 30 10:05:14 2026.
Package wget-1.21.3-1.amzn2023.0.5.x86_64 is already installed.
Package tar-2:1.34-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
================================================================================================================================
 Package                                 Architecture   Version                                 Repository             Size
================================================================================================================================
Installing:
 java-17-amazon-corretto-devel           x86_64         1:17.0.21.0.10-1.amzn2023.1             amazonlinux           142 k
Installing dependencies:
 cairo                                   x86_64         1.18.0-4.amzn2023.0.3                   amazonlinux           717 k
 fonts-filesystem                        noarch         1:2.0.5-12.amzn2023.0.2                 amazonlinux           9.5 k
 google-noto-fonts-common                noarch         20240401-1.amzn2023.0.2                 amazonlinux            17 k
 google-noto-sans-vf-fonts               noarch         20240401-1.amzn2023.0.2                 amazonlinux           593 k
 graphite2                               x86_64         1.3.14-7.amzn2023.0.3                   amazonlinux            97 k
 harfbuzz                                x86_64         7.0.0-2.amzn2023.0.2                    amazonlinux           873 k
 java-17-amazon-corretto-headless        x86_64         1:17.0.21.0.10-1.amzn2023.1             amazonlinux            91 M
 javapackages-filesystem                 noarch         6.0.0-7.amzn2023.0.6                    amazonlinux            12 k
 langpacks-core-font-en                  noarch         3.0-21.amzn2023.0.4                     amazonlinux            10 k
 libX11                                  x86_64         1.8.10-2.amzn2023.0.2                   amazonlinux           655 k
 libX11-common                           noarch         1.8.10-2.amzn2023.0.2                   amazonlinux           144 k
 libXau                                  x86_64         1.0.11-6.amzn2023.0.1                   amazonlinux            33 k
 libXext                                 x86_64         1.3.6-1.amzn2023.0.1                    amazonlinux            42 k
 libXrender                              x86_64         0.9.11-6.amzn2023.0.2                   amazonlinux            28 k
 libbrotli                               x86_64         1.1.0-1.amzn2023.0.2                    amazonlinux           333 k
 libjpeg-turbo                           x86_64         2.1.4-2.amzn2023.0.5                    amazonlinux           190 k
```

```
[ec2-user@ip-192-168-0-84 ~]$ mkdir -p /opt/tomcat
mkdir: cannot create directory '/opt/tomcat': Permission denied
[ec2-user@ip-192-168-0-84 ~]$ sudo -i
[root@ip-192-168-0-84 ~]# mkdir -p /opt/tomcat
[root@ip-192-168-0-84 ~]# cd /opt/tomcat
[root@ip-192-168-0-84 tomcat]# cd /tmp
wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
sudo tar -xf apache-tomcat-10.1.31.tar.gz -C /opt/tomcat --strip-components=1
--2026-10-30 10:15:53--  https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-10-30 10:15:53 ERROR 404: Not Found.

tar: apache-tomcat-10.1.31.tar.gz: Cannot open: No such file or directory
tar: Error is not recoverable: exiting now
[root@ip-192-168-0-84 tmp]# cd /opt/tomcat
[root@ip-192-168-0-84 tomcat]# sudo tar -xf apache-tomcat-10.1.31.tar.gz -C /opt/tomcat --strip-components=1
tar: apache-tomcat-10.1.31.tar.gz: Cannot open: No such file or directory
tar: Error is not recoverable: exiting now
[root@ip-192-168-0-84 tomcat]# cd /tmp
wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
sudo tar -xf apache-tomcat-10.1.31.tar.gz -C /opt/tomcat --strip-components=1
--2026-10-30 10:17:21--  https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-10-30 10:17:21 ERROR 404: Not Found.

tar: apache-tomcat-10.1.31.tar.gz: Cannot open: No such file or directory
tar: Error is not recoverable: exiting now
[root@ip-192-168-0-84 tmp]# cd ..
[root@ip-192-168-0-84 /]# cd /opt/tomcat
[root@ip-192-168-0-84 tomcat]# wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
```

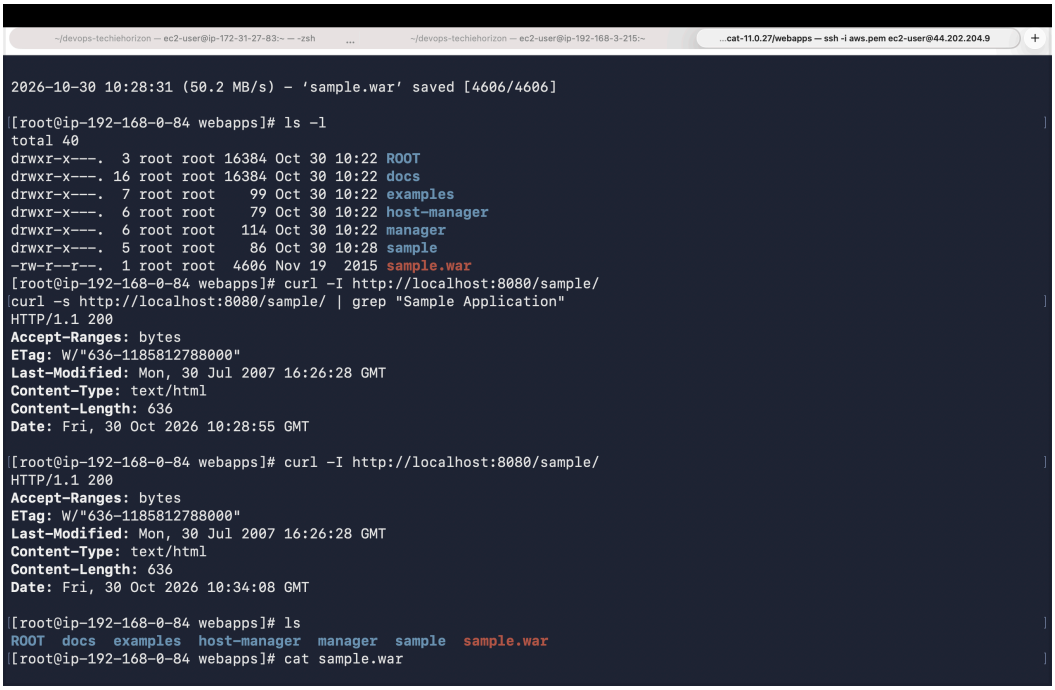
```
[root@ip-192-168-0-84 tomcat]# wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
sudo tar -xf apache-tomcat-10.1.31.tar.gz -C /opt/tomcat --strip-components=1
--2026-10-30 10:17:50--  https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-10-30 10:17:51 ERROR 404: Not Found.

tar: apache-tomcat-10.1.31.tar.gz: Cannot open: No such file or directory
tar: Error is not recoverable: exiting now
[root@ip-192-168-0-84 tomcat]# wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
--2026-10-30 10:19:03--  https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-10-30 10:19:03 ERROR 404: Not Found.

[root@ip-192-168-0-84 tomcat]# wget -o https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.
z
wget: https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz: No such file or directory
[root@ip-192-168-0-84 tomcat]# curl https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.31/bin/apache-tomcat-10.1.31.tar.gz
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
</body></html>
[root@ip-192-168-0-84 tomcat]# wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.27/bin/apache-tomcat-11.0.27.tar.gz
--2026-10-30 10:21:30--  https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.27/bin/apache-tomcat-11.0.27.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
```



```
```



```
[root@ip-192-168-0-84 tomcat]# cd apache-tomcat-11.0.27/bin
[root@ip-192-168-0-84 bin]# ls
bootstrap.jar       ciphers.sh                  daemon.sh     migrate.bat       shutdown.sh          tool-wrapper.bat
catalina-tasks.xml  commons-daemon-native.tar.gz digest.bat    migrate.sh        startup.bat          tool-wrapper.sh
catalina.bat        commons-daemon.jar          digest.sh     setclasspath.bat  startup.sh           version.bat
catalina.sh         configtest.bat              makebase.bat  setclasspath.sh   tomcat-juli.jar      version.sh
ciphers.bat         configtest.sh               makebase.sh   shutdown.bat      tomcat-native.tar.gz
[root@ip-192-168-0-84 bin]# chmod 755  startup.sh
[root@ip-192-168-0-84 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat/apache-tomcat-11.0.27
Using CATALINA_HOME:   /opt/tomcat/apache-tomcat-11.0.27
Using CATALINA_TMPDIR: /opt/tomcat/apache-tomcat-11.0.27/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/apache-tomcat-11.0.27/bin/bootstrap.jar:/opt/tomcat/apache-tomcat-11.0.27/bin/tomcat-j
uli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-192-168-0-84 bin]# curl -I http://localhost:8080/
HTTP/1.1 200
Content-Type: text/html;charset=UTF-8
Date: Fri, 30 Oct 2026 10:27:51 GMT

[root@ip-192-168-0-84 bin]# cd ../webapps
wget https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war
--2026-10-30 10:28:31--  https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'
```



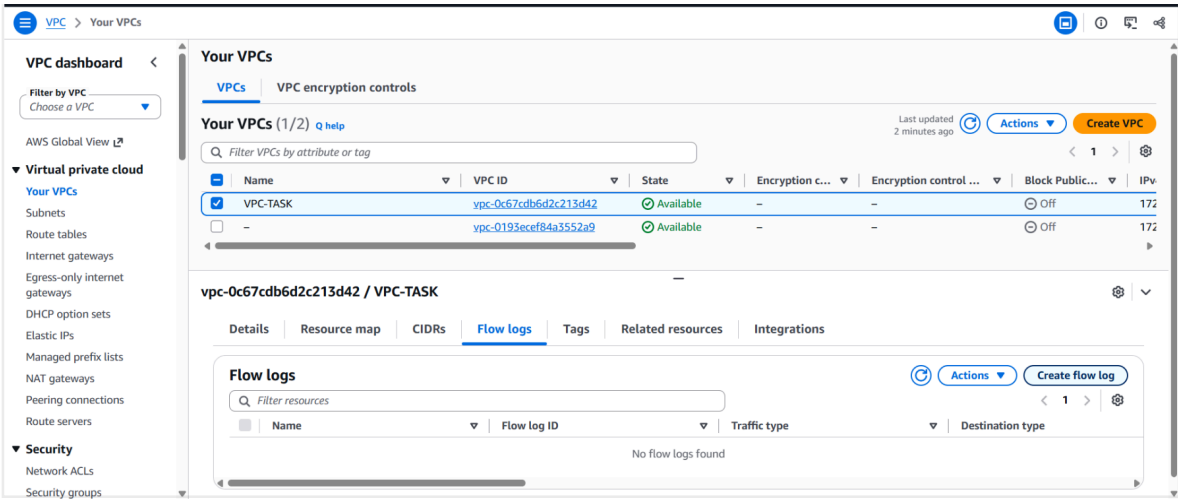
```
~/devops-techiehorizon — ec2-user@ip-172-31-27-83:~ — -zsh   ...   ~/devops-techiehorizon — ec2-user@ip-192-168-3-215:~   ...cat-11.0.27/webapps — ssh -i aws.pem ec2-user@44.202.204.9   +

2026-10-30 10:28:31 (50.2 MB/s) - 'sample.war' saved [4606/4606]

[root@ip-192-168-0-84 webapps]# ls -l
total 40
drwxr-x---.  3 root root 16384 Oct 30 10:22 ROOT
drwxr-x---. 16 root root 16384 Oct 30 10:22 docs
drwxr-x---.  7 root root    99 Oct 30 10:22 examples
drwxr-x---.  6 root root    79 Oct 30 10:22 host-manager
drwxr-x---.  6 root root   114 Oct 30 10:22 manager
drwxr-x---.  5 root root    86 Oct 30 10:28 sample
-rw-r--r--.  1 root root  4606 Nov 19  2015 sample.war
[root@ip-192-168-0-84 webapps]# curl -I http://localhost:8080/sample/
curl -s http://localhost:8080/sample/ | grep "Sample Application"
HTTP/1.1 200
Accept-Ranges: bytes
ETag: W/"636-1185812788000"
Last-Modified: Mon, 30 Jul 2007 16:26:28 GMT
Content-Type: text/html
Content-Length: 636
Date: Fri, 30 Oct 2026 10:28:55 GMT

[root@ip-192-168-0-84 webapps]# curl -I http://localhost:8080/sample/
HTTP/1.1 200
Accept-Ranges: bytes
ETag: W/"636-1185812788000"
Last-Modified: Mon, 30 Jul 2007 16:26:28 GMT
Content-Type: text/html
Content-Length: 636
Date: Fri, 30 Oct 2026 10:34:08 GMT

[root@ip-192-168-0-84 webapps]# ls
ROOT  docs  examples  host-manager  manager  sample  sample.war
[root@ip-192-168-0-84 webapps]# cat sample.war
```

10.Configure VPC flow logs and store the logs

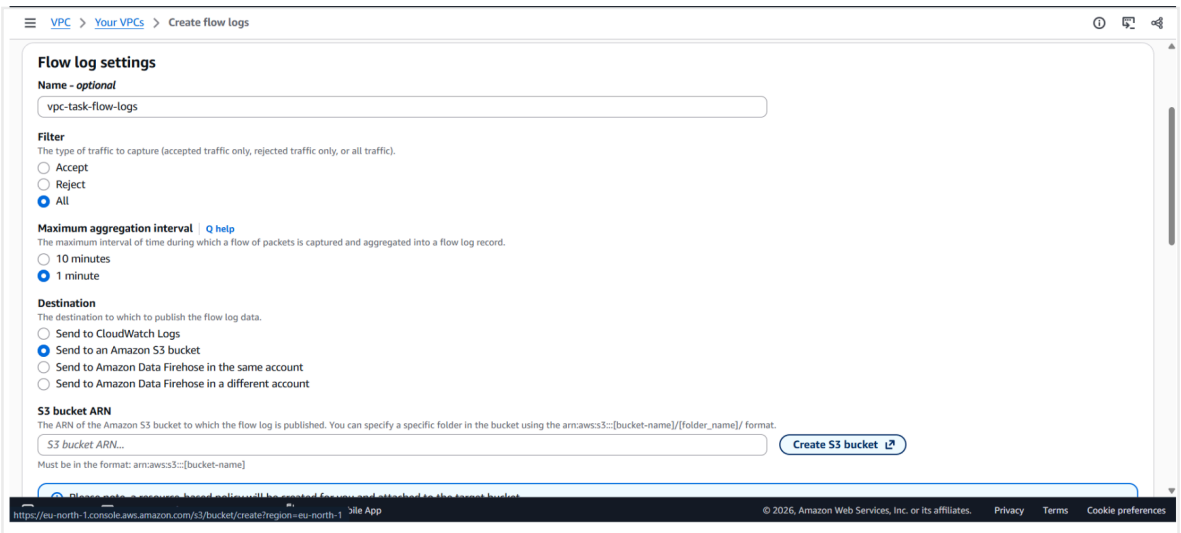
- in S3 and CloudWatch.
- Steps to execute:
- • Navigate to Amazon S3 → Buckets.
- • Click Create bucket and provide a unique bucket name.
- • Select the appropriate AWS Region and keep the recommended bucket settings.
- • Create the S3 bucket for storing VPC Flow Logs.
- • Navigate to CloudWatch → Logs → Log groups.
- • Create a log group named VPC-Flow-Logs.
- • Navigate to VPC → Your VPCs.
- • Select the VPC created in Task 1 and open the Flow Logs tab.
- • Click Create flow log and provide a suitable name.
- • Select All traffic to capture accepted and rejected traffic.

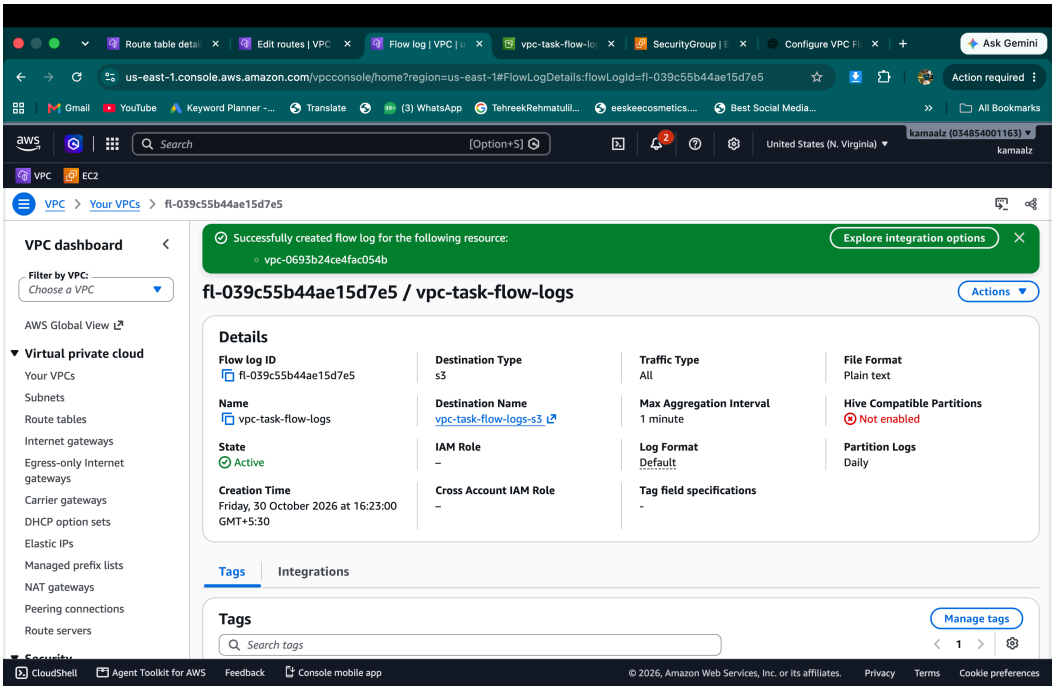
- • Choose the required aggregation interval.
- • Select S3 as the destination and choose the bucket created earlier.
- • Configure the required permissions and create the flow log.
- • Create another flow log for the same VPC.
- • Select CloudWatch Logs as the destination and choose the VPC-Flow-Logs log group.
- • Create a new service role or select an existing role with the required CloudWatch permissions.
- • Create the flow log and verify that its state is Active.
- • Navigate to the S3 bucket and open the AWSLogs folder.
- • Navigate through the generated folders and locate the .log.gz files.
- • Download one log file, extract it using 7-Zip, and inspect the traffic records.

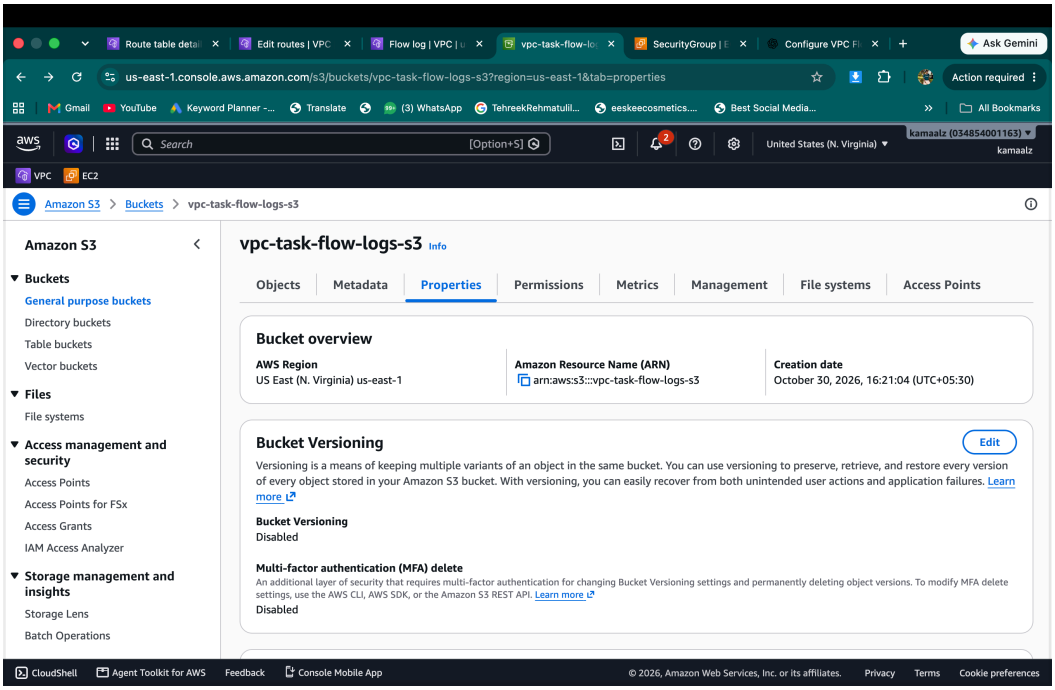
1. • Open CloudWatch Logs and verify that the flow log records are available in the log group.

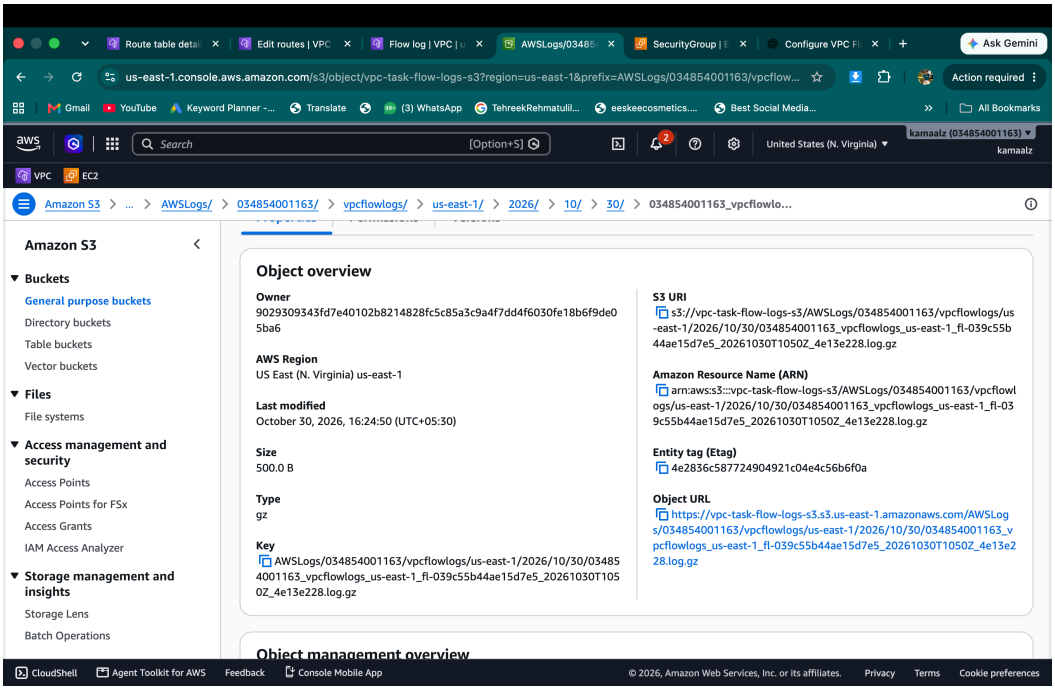


2.

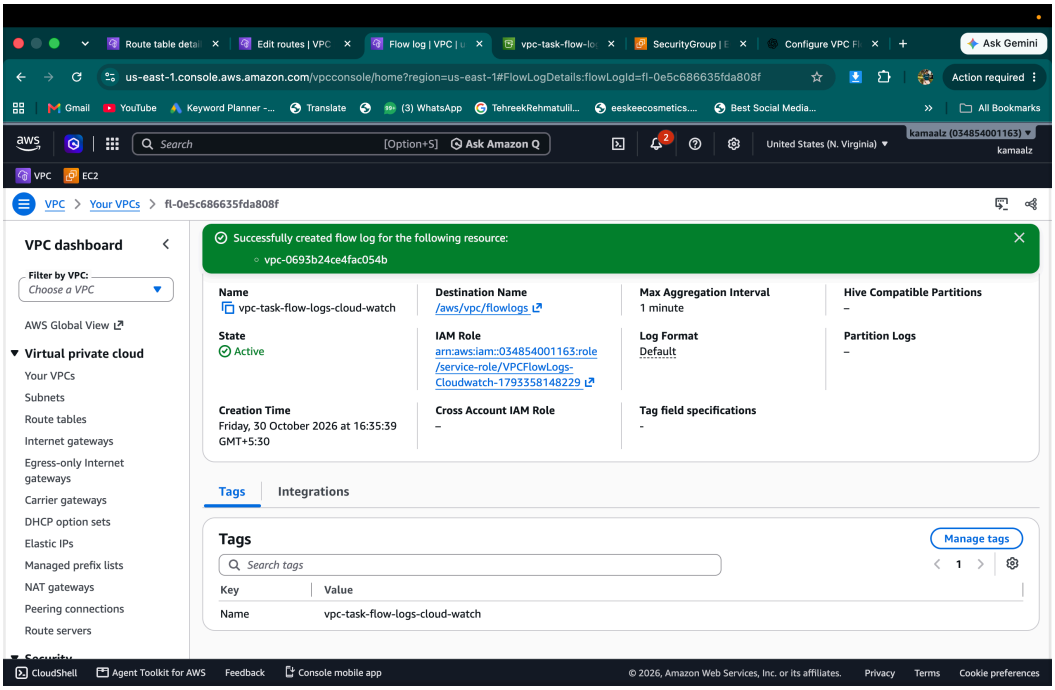


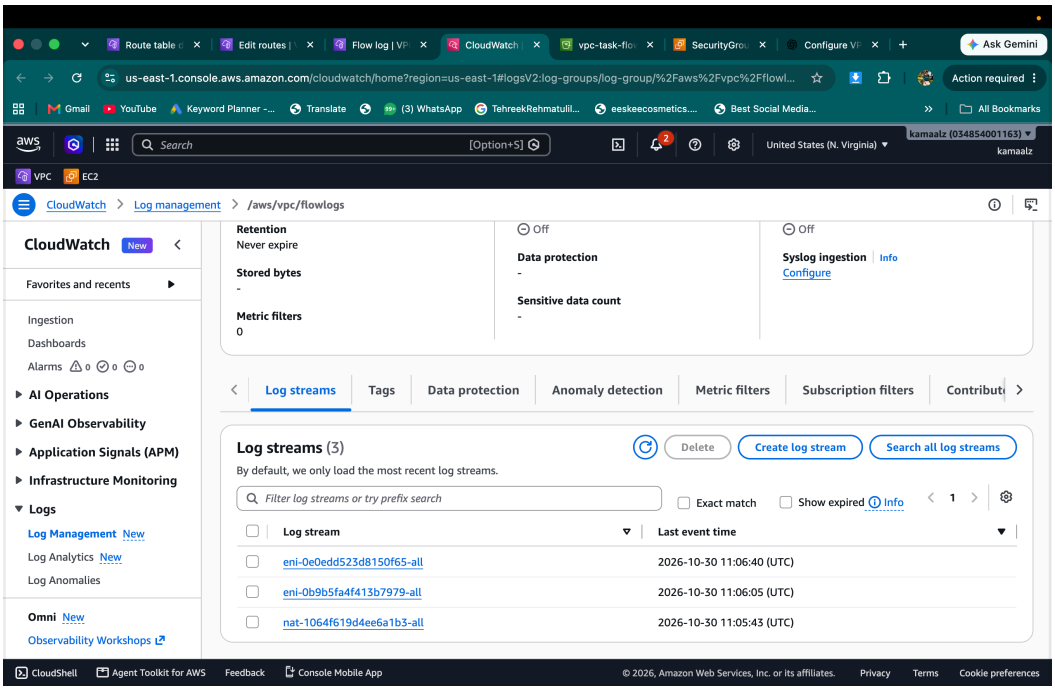

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#FlowLogDetails:flowLogId=fl-039c55b44ae15d7e5
VPC > Your VPCs > fl-039c55b44ae15d7e5
VPC dashboard
Successfully created flow log for the following resource:
vpc-0693b24ce4fac054b
Explore integration options
fl-039c55b44ae15d7e5 / vpc-task-flow-logs
Actions
Details
Flow log ID
fl-039c55b44ae15d7e5
Destination Type
s3
Traffic Type
All
File Format
Plain text
Name
vpc-task-flow-logs
Destination Name
vpc-task-flow-logs-s3
Max Aggregation Interval
1 minute
Hive Compatible Partitions
Not enabled
State
Active
IAM Role
–
Log Format
Default
Partition Logs
Daily
Creation Time
Friday, 30 October 2026 at 16:23:00 GMT+5:30
Cross Account IAM Role
–
Tag field specifications
-
Tags
Integrations
Tags
Manage tags
Search tags


us-east-1.console.aws.amazon.com/s3/buckets/vpc-task-flow-logs-s3?region=us-east-1&tab=properties
Amazon S3 > Buckets > vpc-task-flow-logs-s3
Amazon S3
vpc-task-flow-logs-s3 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Bucket overview
AWS Region
US East (N. Virginia) us-east-1
Amazon Resource Name (ARN)
arn:aws:s3:::vpc-task-flow-logs-s3
Creation date
October 30, 2026, 16:21:04 (UTC+05:30)
Bucket Versioning
Edit
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disabled
Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more
Disabled


us-east-1.console.aws.amazon.com/s3/object/vpc-task-flow-logs-s3?region=us-east-1&prefix=AWSLogs/034854001163/vpcflow...
Amazon S3 > ... > AWSLogs/ > 034854001163/ > vpcflowlogs/ > us-east-1/ > 2026/ > 10/ > 30/ > 034854001163_vpcflowlo...
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Object overview
Owner
9029309343fd7e40102b8214828fc5c85a3c9a4f7dd4f6030fe18b6f9de05ba6
AWS Region
US East (N. Virginia) us-east-1
Last modified
October 30, 2026, 16:24:50 (UTC+05:30)
Size
500.0 B
Type
gz
Key
AWSLogs/034854001163/vpcflowlogs/us-east-1/2026/10/30/034854001163_vpcflowlogs_us-east-1_fl-039c55b44ae15d7e5_20261030T1050Z_4e13e228.log.gz
S3 URI
s3://vpc-task-flow-logs-s3/AWSLogs/034854001163/vpcflowlogs/us-east-1/2026/10/30/034854001163_vpcflowlogs_us-east-1_fl-039c55b44ae15d7e5_20261030T1050Z_4e13e228.log.gz
Amazon Resource Name (ARN)
arn:aws:s3:::vpc-task-flow-logs-s3/AWSLogs/034854001163/vpcflowlogs/us-east-1/2026/10/30/034854001163_vpcflowlogs_us-east-1_fl-039c55b44ae15d7e5_20261030T1050Z_4e13e228.log.gz
Entity tag (Etag)
4e2836c587724904921c04e4c56b6f0a
Object URL
https://vpc-task-flow-logs-s3.s3.us-east-1.amazonaws.com/AWSLogs/034854001163/vpcflowlogs/us-east-1/2026/10/30/034854001163_vpcflowlogs_us-east-1_fl-039c55b44ae15d7e5_20261030T1050Z_4e13e228.log.gz
Object management overview
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

cloud watch


us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#FlowLogDetails:flowLogId=fl-0e5c686635fda808f
VPC > Your VPCs > fl-0e5c686635fda808f
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Successfully created flow log for the following resource:
vpc-0693b24ce4fac054b
Name
vpc-task-flow-logs-cloud-watch
Destination Name
/aws/vpc/flowlogs
Max Aggregation Interval
1 minute
Hive Compatible Partitions
–
State
Active
IAM Role
arn:aws:iam::034854001163:role/service-role/VPCFlowLogs-Cloudwatch-1793358148229
Log Format
Default
Partition Logs
–
Creation Time
Friday, 30 October 2026 at 16:35:39 GMT+5:30
Cross Account IAM Role
–
Tag field specifications
-
Tags
Integrations
Tags
Manage tags
Search tags
Key
Value
Name
vpc-task-flow-logs-cloud-watch
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Route table
Edit routes
Flow log | VP
CloudWatch
vpc-task-flo
SecurityGro
Configure VP
Ask Gemini
us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2:log-groups/log-group/%2Faws%2Fvpc%2Fflowl...
Action required
Gmail
YouTube
Keyword Planner -...
Translate
(3) WhatsApp
TehreekRehmatulil...
eeskeecosmetics....
Best Social Media...
All Bookmarks
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
VPC
EC2
CloudWatch
Log management
/aws/vpc/flowlogs
CloudWatch
New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
New
Log Analytics
New
Log Anomalies
Omni
New
Observability Workshops
Retention
Never expire
Stored bytes
-
Metric filters
0
Off
Data protection
-
Sensitive data count
-
Off
Syslog ingestion
Info
Configure
Log streams
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Contribute
Log streams (3)
By default, we only load the most recent log streams.
Delete
Create log stream
Search all log streams
Filter log streams or try prefix search
Exact match
Show expired
Info
1
Log stream
Last event time
eni-0e0edd523d8150f65-all
2026-10-30 11:06:40 (UTC)
eni-0b9b5fa4f413b7979-all
2026-10-30 11:06:05 (UTC)
nat-1064f619d4ee6a1b3-all
2026-10-30 11:05:43 (UTC)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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